



BHARAT COLLEGE OF ENGINEERING

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EXPERIMENT NO-08

AIM: Write a C program to demonstrate the concept of Page replacement policies for handling page faults e.g. FIFO, LRU.

THEORY: In an operating system that uses paging, a page replacement algorithm is needed when a page fault occurs and no free page frame is available. In this case, one of the existing pages in memory must be replaced with the new page.

The efficiency of a page replacement algorithm directly affects the page fault rate, which in turn impacts system performance.

Common Page Replacement Techniques

- First In First Out (FIFO)
- Optimal Page replacement
- Least Recently Used (LRU)
- Most Recently Used (MRU)

1. First In First Out (FIFO)

This is the simplest page replacement algorithm. In this algorithm, the operating system keeps track of all pages in the memory in a queue, the oldest page is in the front of the queue. When a page needs to be replaced page in the front of the queue is selected for removal.

Example 1: Consider page reference string 1, 3, 0, 3, 5, 6, 3 with 3-page frames. Find the number of page faults using FIFO Page Replacement Algorithm.



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Page
reference

1, 3, 0, 3, 5, 6, 3

1	3	0	3	5	6	3
		0	0	0	0	3
	3	3	3	3	6	6
1	1	1	1	5	5	5
Miss	Miss	Miss	Hit	Miss	Miss	Miss

Total Page Fault = 6

- Initially, all slots are empty, so when 1, 3, 0 came they are allocated to the empty slots ---> **3 Page Faults.**
- When 3 comes, it is already in memory so ---> **0 Page Faults.**
- Then 5 comes, it is not available in memory, so it replaces the oldest page slot i.e 1. ---> **1 Page Fault.**
- 6 comes, it is also not available in memory, so it replaces the oldest page slot i.e 3 ---> **1 Page Fault.**
- Finally, when 3 come it is not available, so it replaces 0 1-page **fault.**

2. Least Recently Used

In this algorithm, page will be replaced which is least recently used.

Example Consider the page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with 4-page frames. Find number of page faults using LRU Page Replacement Algorithm.



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Page
reference

7,0,1,2,0,3,0,4,2,3,0,3,2,3

No. of Page frame - 4

7	0	1	2	0	3	0	4	2	3	0	3	2	3
			2	2	2	2	2	2	2	2	2	2	2
		1	1	1	1	1	4	4	4	4	4	4	4
	0	0	0	0	0	0	0	0	0	0	0	0	0
7	7	7	7	7	3	3	3	3	3	3	3	3	3
Miss	Miss	Miss	Miss	Hit	Miss	Hit	Miss	Hit	Hit	Hit	Hit	Hit	Hit

Total Page Fault = 6

- Initially, all slots are empty, so when 7 0 1 2 are allocated to the empty slots ---> 4 Page faults.
- 0 is already there so ---> 0 Page fault. when 3 came it will take the place of 7 because it is least recently used ---> 1 Page fault.
- 0 is already in memory so ---> 0 Page fault.
- 4 will takes place of 1 ---> 1 Page Fault.
- Now for the further page reference string ---> 0 Page fault because they are already available in the memory.

PROGRAM:

FIFO:

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
int i,j,n,a[50],frame[10],no,k,avail,count=0;
```

```
printf("\n ENTER THE NUMBER OF PAGES:\n");
```

```
scanf("%d",&n);
```



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```
printf("\n ENTER THE PAGE NUMBER :\n");

for(i=1;i<=n;i++)

scanf("%d",&a[i]);

printf("\n ENTER THE NUMBER OF FRAMES :");

scanf("%d",&no);

for(i=0;i<no;i++)

    frame[i]= -1;

    j=0;

    printf("\tref string\t page frames\n");

for(i=1;i<=n;i++)

{

    printf("%d\t\t",a[i]);

    avail=0;

    for(k=0;k<no;k++)

if(frame[k]==a[i])

        avail=1;

        if (avail==0)

        {

            frame[j]=a[i];

            j=(j+1)%no;

            count++;

            for(k=0;k<no;k++)

                printf("%d\t",frame[k]);

        }

}
```



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```

printf("\n");
}

printf("Page Fault Is %d",count);

return 0;
}

```

OUTPUT:

ENTER THE NUMBER OF PAGES: 20

ENTER THE PAGE NUMBER : 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

ENTER THE NUMBER OF FRAMES :3

	ref string	page frames
7	7	-1 -1
0	7	0 -1
1	7	0 1
2	2	0 1
0		
3	2	3 1
0	2	3 0
4	4	3 0
2	4	2 0
3	4	2 3
0	0	2 3
3		
2		



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1 0 1 3

2 0 1 2

0

1

7 7 1 2

0 7 0 2

1 7 0 1

Page Fault Is 15

LRU:

```
#include <stdio.h>
```

```
int main() {
```

```
    int frames, pages, i, j, k, pos, fault = 0;
```

```
    printf("Enter number of frames: ");
```

```
    scanf("%d", &frames);
```

```
    printf("Enter number of pages: ");
```

```
    scanf("%d", &pages);
```

```
    int ref[pages], frame[frames], time[frames];
```



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```
printf("Enter reference string: ");
```

```
for(i = 0; i < pages; i++) {  
    scanf("%d", &ref[i]);  
}
```

```
// Initialize frames
```

```
for(i = 0; i < frames; i++) {  
    frame[i] = -1;  
}
```

```
for(i = 0; i < pages; i++) {
```

```
    int found = 0;
```

```
    // Check if page is already in frame
```

```
    for(j = 0; j < frames; j++) {
```

```
        if(frame[j] == ref[i]) {
```

```
            found = 1;
```

```
            time[j] = i; // Update recent use
```

```
            break;
```

```
        }
```

```
    }
```

```
// Page fault
```

```
if(found == 0) {
```



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```
pos = 0;
```

```
// Find least recently used
```

```
for(j = 1; j < frames; j++) {
```

```
    if(time[j] < time[pos]) {
```

```
        pos = j;
```

```
    }
```

```
}
```

```
frame[pos] = ref[i];
```

```
time[pos] = i;
```

```
fault++;
```

```
printf("\nPage Fault: ");
```

```
} else {
```

```
    printf("\nPage Hit: ");
```

```
}
```

```
// Print frame status
```

```
for(k = 0; k < frames; k++) {
```

```
    if(frame[k] != -1)
```

```
        printf("%d ", frame[k]);
```

```
    else
```

```
        printf("- ");
```




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}

}

```
printf("\n\nTotal Page Faults = %d\n", fault);
```

```
return 0;
```

```
}
```

OUTPUT:

Page Fault: 1 - -

Page Fault: 1 2 -

Page Fault: 1 2 3

Page Fault: 4 2 3

Page Fault: 4 1 3

Page Fault: 4 1 2

Page Fault: 5 1 2

Page Hit: 5 1 2

Page Hit: 5 1 2

Page Fault: 3 1 2

Page Fault: 3 4 2

Page Fault: 3 4 5

Total Page Faults = 10